LEH stability — benchmark report

Agent-alone candidates, published prediction values and reference variants calculated in this work.

Agent Scientist: gpt-6.1-sol. Predictor names and units are listed with each measurement.

Published predictions and our predictions

Property / predictor	Published prediction	Our candidates
Predicted stability change	FoldX: T85I −20; T85V −14 kJ/mol ΔΔGfold	DDGun_SEQ: LEH_R004 +0.2; LEH_R003 +0.5 kcal/mol ΔΔGunfold
Predicted Tm change (°C)	Not reported	LEH_R004 +1.2498; LEH_R003 +0.3552

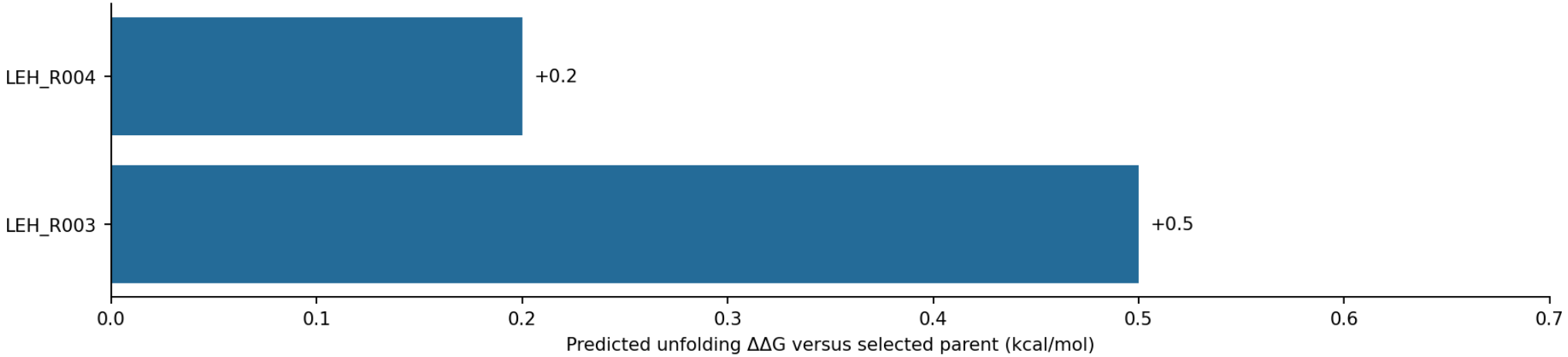
Source: Wijma et al., Figure 1 and methods

Our stability predictions

Candidate	Mutations	DDGun unfolding ΔΔG (kcal/mol)
LEH_R004	K42R,P107A,E124D	0.2
LEH_R003	K42Y,P107A,E124D	0.5

Change is relative to our selected parent. Positive DDGun unfolding ΔΔG indicates predicted stabilization; negative indicates predicted destabilization.

Stability change from our parent

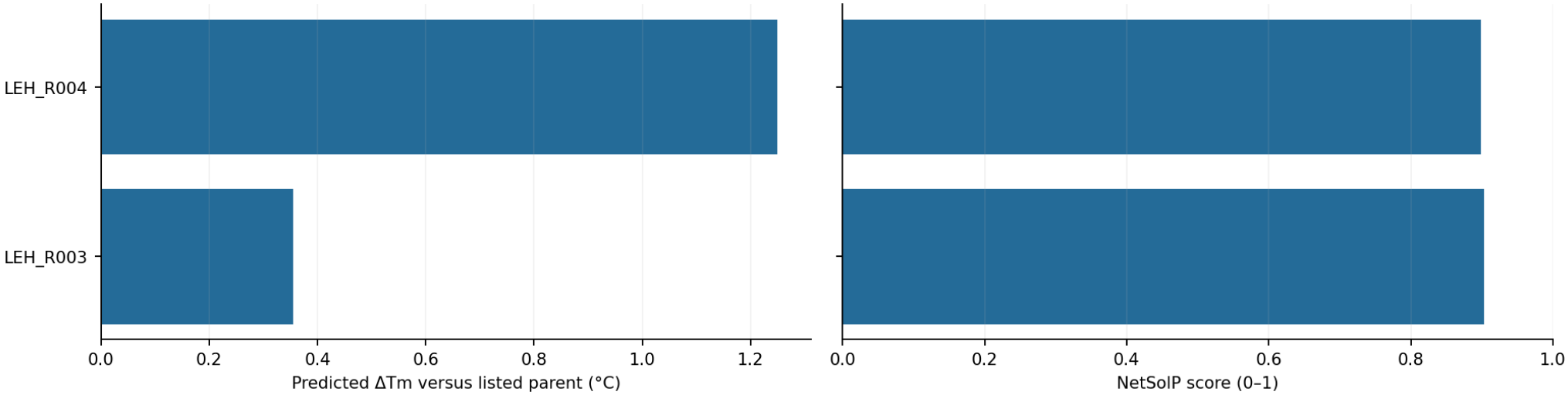


All bars use DDGun_SEQ and the same selected parent within this task.

Property predictions calculated in this work

Sequence	Group	Parent / background	Predicted Tm (°C)	ΔTm (°C)	NetSolP	pHPred (raw)
LEH_R004	Our candidate	LEH_WT	53.4325	1.2498	0.898987	7.3929
LEH_R003	Our candidate	LEH_WT	52.5379	0.3552	0.902987	7.3786

Candidate and reference property predictions



Blue: our candidates. Amber: reference variants calculated in this work. Changes use the parents listed in the table.

Parent identity, preparation details and run records: results/task_002_leh_stability/agent-alone/README.md.